

# MOSAM BISWAS

REVISED · SEPTEMBER 2026

The complete technical anatomy of [www.mosambiswas.com](http://www.mosambiswas.com), a dark editorial portfolio built with zero frameworks and zero build step. Version six set the whole site on fire: a hand written WebGL ember field, an inertial motion system, three page specific choreographies. This revision adds what came after the fire: an *offline first* application that carries the entire site, type included, onto a phone with no network at all.

(01) · OVERVIEW

# ONE REPO, NO *machinery*

The site is a single static repository, `BiswasMosam.github.io`, served straight from GitHub Pages at `www.mosambiswas.com`. There is no bundler, no transpiler, no `package.json`. Every line that ships is a line that was written by hand: four HTML pages, two stylesheets and eight small scripts.

**What Vol. 06 brought.** Three motion engines joined the v5 base: `shader.js` (a full page WebGL ember field), `motion.js` (inertial scroll, velocity shear, letter cascades), and two page specific systems, `cert-motion.js` for the certificates dossier and `sheichobi/darkroom.js` for the photography darkroom.

**What this revision adds.** The site stopped being only a website. It is now also an installable application that precaches itself in full, a résumé page in the same design system, and a contact form that refuses to lie about whether your message left. Those three are documented in sections 08 and 09.

COUNTED, NOT ESTIMATED

2,192 lines of JavaScript across eight files, 0 dependencies, 0 build steps. The entire site is what you see in the repo.

| FILE              | ROLE                                                                                                       |
|-------------------|------------------------------------------------------------------------------------------------------------|
| main.js           | Preloader gate, cursor and bubbles, reveals, parallax, clock, menu, copy to clipboard, contact form, modal |
| motion.js         | Inertial scroll, shear, cascades, marquee, magnetic buttons                                                |
| shader.js         | WebGL ember field, GLSL fragment shader                                                                    |
| cert-motion.js    | Dossier motion for certificates.html                                                                       |
| darkroom.js       | Darkroom motion for the SheiChobi gallery                                                                  |
| sheichobi.js      | Gallery: filters, load more, lightbox, collections                                                         |
| service-worker.js | REV Offline first cache. Precaches the whole shell at install                                              |
| sw-register.js    | NEW Registration, update flow, dead origin eviction                                                        |
| style.css         | Design system and every page level style                                                                   |
| fonts.css         | NEW Self hosted @font-face, latin subsets                                                                  |
| resume.html       | NEW Web résumé in the site's design system                                                                 |

FOUR PAGES

`index.html` (portfolio) · `resume.html` · `certificates.html` · `sheichobi/sheichobi.html` (photography). All four are precached, so all four work offline.

"No frameworks, no build step. Every line that ships was written *by hand*."

(02) · DESIGN SYSTEM

# WARM BLACK, BONE INK, *vermilion*

The palette is deliberately narrow: a warm near black canvas, an off white "bone" ink, and exactly one accent. Hierarchy is carried by two dimmed alpha steps of the ink rather than extra hues. Hairlines at 14% ink alpha do all the structural work, the same rule that frames this page.

The editorial voice comes from numbered section labels, (01) · SELECTED WORK, set in Space Mono, and from emphasised words dropped into *Instrument Serif italic* in accent colour, mid sentence, exactly as on this page.

## HOUSE RULES

Easing: `cubic-bezier(0.22, 1, 0.36, 1)`, a long decelerating tail used by nearly every transition. Fluid padding: `clamp(1.25rem, 4vw, 4rem)`.

### COPY RULE

No em or en dashes anywhere in visible copy. Ranges read 2022 / 2026, separators are middots, and inline dashes became commas, colons or full stops. Ordinary hyphens in compounds stay. This document follows the same rule.

| TOKEN       | VALUE   | USE                        |
|-------------|---------|----------------------------|
| --bg        | #0b0b0a | Warm black canvas          |
| --bg-2      | #131311 | Raised surfaces            |
| --ink       | #edebe4 | Primary text               |
| --ink-dim   | ink 58% | Body copy                  |
| --ink-faint | ink 34% | Meta text                  |
| --line      | ink 14% | Hairline rules             |
| --accent    | #ff5227 | Vermilion, the only colour |

## TYPE, NOW SELF HOSTED REV

The four faces used to come from Google Fonts over the network. They are now served from `/fonts/` as latin subset woff2 files declared in `fonts.css`, because an offline application that falls back to Arial is not the same site. That change is what makes the install in section 08 honest.

| FACE                           | ROLE                        |
|--------------------------------|-----------------------------|
| Syne 600/700/800               | Display, wordmark, headings |
| Manrope 400/500/600            | Body copy                   |
| Space Mono 400                 | Labels, meta, numerals      |
| Instrument Serif 400 <i>it</i> | Accent words, mid sentence  |

## ONE LANGUAGE, FOUR DIALECTS

The tokens above are shared by every page without exception. What differs is the **motion vocabulary**: each page borrows a metaphor from the thing it is showing you. The rest of this document is organised around that split.

| PAGE              | DIALECT  | GATE         |
|-------------------|----------|--------------|
| index.html        | Embers   | has-motion   |
| certificates.html | Dossier  | has-dossier  |
| sheichobi.html    | Darkroom | has-darkroom |
| resume.html       | Print    | @media print |

### WHY GATES AND NOT SCRIPTS

Each dialect is a class its script puts on `<html>`, and every rule for it is written behind that class. A page with JavaScript disabled, or a visitor who asked for reduced motion, gets a stylesheet where those blocks simply never match. Nothing has to be undone, because *nothing was ever applied*.

## (03) · THE HERO

# EDGE TO EDGE, TO THE *pixel*

The hero name **MOSAM / BISWAS** is fitted to the exact container width by JavaScript rather than by a fixed type scale: each line is set at a 100px probe size, measured, then scaled by `containerWidth / wordWidth`. The second line is outlined via `-webkit-text-stroke` with a transparent fill.

**Why one pass is not enough.** Pixel based letter spacing and stroke widths do not scale with font size, so a single corrective pass always lands slightly wide. `fitLines()` runs up to three passes and breaks early once the measured width is within 1px of the true edge.

Because `motion.js` splits the hero into per letter spans after the first fit, it exposes the fitter as `window.__refitHero` and calls it again once the split is done. Resize is debounced onto the same function.

```
// main.js · fit each display line to its box
let size = 100;
for (let pass = 0; pass < 3; pass++) {
  const width = word.getBoundingClientRect().width;
  if (width <= 0) break;
  size *= line.clientWidth / width;
  line.style.fontSize = `${size.toFixed(2)}px`;
  if (Math.abs(width - line.clientWidth) < 1) break;
}

// motion.js re fits after the letter split
window.__refitHero = fitLines;
```

**DETAIL**

The same measure and scale trick fits the **LET'S TALK** contact title and the SheiChobi and certificates hero lines. One function, four pages.

**THE SECTION HEAD GRAMMAR**

Every section on every page opens with the same three part header, which is what makes four differently animated pages read as one document.

`--label` Space Mono, accent, (01) ·  
Selected Work

`--title` Display, with one word in serif  
italic

`--aside` Mono meta, a count or a range

Ranges read **2022 / 2026**, never with a dash. On narrow screens `.sec-head--stack` drops the aside under the title instead of letting it collide.

**WHAT THE GRID DROPS**

The work list is a CSS grid of five columns: index, title, category, meta and arrow. Rather than reflowing into cards, it **sheds columns** as the viewport narrows, in order of how much each one earns its space.

| BELOW  | DROPS                                      |
|--------|--------------------------------------------|
| 1100px | Category line                              |
| 900px  | Tech stack meta                            |
| Always | Index, title and arrow survive every width |

That ordering has a consequence worth stating: **the title is the only column present at every width**, so any per project marker has to live there. It is why the **Live** badge is a `<sup>` on the title and not a column of its own.

**MASKS, NOT OPACITY**

Display lines sit inside **overflow: hidden** wrappers, so a heading can rise into place from below the baseline rather than fading in. The mask is part of the markup from the start, which means the same structure serves the v5 reveal, the v6 letter cascade and the dossier redaction sweep without any of them knowing about the others.

**WHY IT IS MEASURED, NOT TYPED**

A fixed `clamp()` type scale would set the hero to a size that is nearly right at every width. Measuring sets it to the size that is *exactly* right at this one. The cost is three `getBoundingClientRect` calls on load and on resize.

(04) · THE EMBER FIELD

# HAND WRITTEN GLSL, NO *libraries*

Behind every page sits a fixed `<canvas>` at `z-index: -1` running a raw WebGL fragment shader. No Three.js, no helper library, just a compiled program and a fullscreen triangle, three vertices, cheaper than a quad. The body background went transparent so the field shows through everything.

The image is **domain warped smoke**: a hash based value noise is stacked into four octaves of `fbm`, then fed into itself twice, `fbm(p + 2.1 · r)` where `r` is built from `q`, which is built from `p`. Bone coloured smoke rises from the folds; a smoothstep ridge extractor paints *vermilion embers* along the crests; a vignette pulls the edges back to editorial black; a one bit dither kills gradient banding.

## THE ENERGY SYSTEM

The field breathes with the page. A single `uEnergy` uniform blends four inputs: hero presence (strong at the top), contact flare (strong at LET'S TALK), smoothed scroll velocity, and a `uBurst` ignition that fires at 1.4 when the preloader lifts and decays by  $\times 0.982$  per frame. Mid read, the field goes quiet.

## CURSOR WARMTH

The mouse position is lerped (0.07) into a `uMouse` uniform; a Gaussian falloff  $\exp(-d^2 \cdot 7)$  adds local heat, scaled by smoothed cursor speed. The warp itself is bent toward the pointer, so the smoke *leans into your hand*.

```
// domain warped fbm · the smoke
vec2 q = vec2(fbm(p*1.7 + t),
              fbm(p*1.7 - t*0.8 + 4.7));
vec2 r = vec2(fbm(p*1.7 + 2.3*q + ... + t*0.55),
              fbm(p*1.7 + 2.3*q + ... - t*0.4));

// the field leans toward the cursor
r -= (p - m) * exp(-md * 2.6) * 0.4;
float f = fbm(p*1.7 + 2.1*r);

// bone smoke + vermilion embers
col += vec3(.93,.92,.89) * f*f*f * 0.13
      * (0.35 + 0.65 * energy);
float ridge = smoothstep(0.50, 0.86,
                        f * (0.55 + 0.45 * q.x));
col += vec3(1.0, 0.32, 0.15) * ridge
      * 0.42 * energy;
```

## PERFORMANCE DISCIPLINE

- **0.5× render scale**, capped at 1.5 dpr. The field is soft by nature, so nobody can tell.
- **Context flags** `alpha:false, antialias:false, depth:false, powerPreference:'low-power'`.
- **Adaptive governor** counts frames over 27ms; after 45 slow frames it degrades resolution once more by  $\times 0.65$ , permanently.
- **Never a dependency**. Under `prefers-reduced-motion`, or if WebGL or compilation fails, the canvas removes itself and the site falls back to plain `--bg`.

## SIX UNIFORMS, AND WHAT THE PAGE PUTS IN THEM

The entire conversation between the document and the shader happens through six numbers, uploaded once per frame. There is no other channel.

| UNIFORM             | CARRIES                                            |
|---------------------|----------------------------------------------------|
| <code>uRes</code>   | Canvas size in device pixels                       |
| <code>uTime</code>  | Seconds since start, scaled to 0.055 in the shader |
| <code>uMouse</code> | Lerped pointer position, y flipped to match GL     |

| UNIFORM              | CARRIES                                                                             |
|----------------------|-------------------------------------------------------------------------------------|
| <code>uMspd</code>   | Smoothed cursor speed, 0 to 1.2                                                     |
| <code>uEnergy</code> | $0.2 + 0.85 \cdot \max(\text{hero}, \text{tail}) + \text{velocity}$ , eased at 0.04 |
| <code>uBurst</code>  | Set to 1.4 once, on <code>body.is-loaded</code> , then $\times 0.982$ per frame     |

(05) · THE MOTION SYSTEM

# THE PAGE LEANS INTO ITS OWN *momentum*

## INERTIAL SCROLL

On fine pointer devices, wheel events are intercepted (`preventDefault`) and fed into a target value; a RAF loop eases the real scroll position toward it at lerp 0.11. Native jumps, keyboard and scrollbar drags, are detected when the actual position drifts more than 1.5px from the last write, and the system resyncs instead of fighting. Anchor links glide on the same easing. `html.smooth` disables native smooth scroll so the two never conflict.

## VELOCITY SHEAR

Scroll velocity is smoothed (lerp 0.12) and mapped to a `skewY` of at most  $\pm 3.4^\circ$  applied to the hero, section heads, work list, research and photography blocks. The page visibly shears with momentum and *settles when you stop.*

## LETTER AND WORD CASCADES

The hero and contact titles are split into per letter `.hero__ltr` spans that rise from `translateY(118%) rotate(5deg)` inside the existing overflow masks, staggered 50ms per letter via a `--i` custom property. Section titles split into words (70ms stagger); work rows and photo strips enter with 60/90ms staggers under `IntersectionObserver`.

## VELOCITY DRIVEN MARQUEE

The CSS keyframe marquee hands over to JS (`.is-js`): base drift 42px/s, plus a velocity term clamped at  $\pm 860$ . It accelerates as you scroll and reverses when you backtrack, wrapping by modulo on the half width of the duplicated track.

## THE REST OF THE CHOREOGRAPHY

- **Header** hides past 280px when velocity  $> 5$ , returns on any upward motion.
- **Progress bar**, a 2px vermilion hairline scaled `scaleX(y / maxScroll)`.
- **Hero lines drift apart**: MOSAM slides left, BISWAS right, up to 220px as you leave the hero.
- **Magnetic buttons**: CTAs and the menu button lean toward the cursor (0.26x / 0.34y of the offset).
- **Stat counters** count up over 1.2s with cubic ease out, zero padded.
- **Scroll hint** fades out after 80px.

### ONE GATE

Everything sits behind `html.has-motion`, which `motion.js` adds only when `prefers-reduced-motion` is off. With JS disabled or motion reduced, the stylesheet's v6 block is inert and the v5 reveals still work.

*"Hand rolled, no libraries." The header comment of every v6 file.*

(06) · BUBBLES

# A CURSOR THAT WHISPERS & *reveals*

The signature interaction. One fixed `#cursor` div, a 12px white dot in `mix-blend-mode: difference`, lerp follows the pointer (factor 0.22) and morphs into three distinct instruments depending on what it touches. Desktop only (`pointer: fine`), removed entirely under reduced motion.

## WHISPER MODE, WIRED TO SEPARATE WORDS

Any element carrying `data-whisper` morphs the dot into an ink coloured caption pill with Space Mono text. Crucially, whispers are wired to **individual words**: in "Developer, researcher, photographer" each word owns its own whisper, so sweeping the sentence produces a running commentary. A `||` delimiter stores alternate lines that cycle on every re hover, tracked via `data-whisperIdx`.

The pill drifts 26px above the pointer, flips below near the viewport top (`targetY < 60` gives `+34`), clamps at the horizontal edges, and sizes itself to label width + 36px by 32px. Links grow the dot to 52px instead, unless they whisper too.

### REVISED · SEPTEMBER 2026

Whispers moved off the headings themselves and onto **inner spans**. A `<h2>` is a block box, so hovering the empty half of a headline used to fire the pill over blank space. Wrapping the words means the bubble only speaks where there is type to speak over.

## LENS MODE, THE READING LENS

Paragraphs marked `.secret` carry two layers: the visible `.secret__base` and an absolutely positioned `.secret__layer`, alternate text set in accent coloured *Instrument Serif italic*. Over a secret, the dot expands into a 78px radius transparent circle with a vermilion border, and the hidden layer is unmasked through a per frame `clip-path: circle(r at x y)` that tracks the cursor. Move the lens, read the subtext.

```
layer.style.clipPath =  
  `circle(${r}px at ${x}px ${y}px)`;  
// lens radius lerp 0.16 · pill offset 0.2
```

## WHERE IT SPEAKS

- **The version label**: "the smoke is hand written GLSL" and "no libraries, just math and embers". The site annotating its own shader.
- **The copyright line**, four rotating variants, including "inspect element is not a license".
- **The photo strip**, one word per frame: street, portrait, wildlife, monochrome, concert.
- **On SheiChobi** the bubble takes `.is-photo` (opacity 0) over photographs, so the difference blend never inverts an image.

The same lerped cursor position that drives the pill also feeds the shader's `uMouse`, so the smoke and the bubble move as *one hand*.

(07) · PRELOADER & LIFECYCLE

# HELD AT 99 UNTIL IT'S *true*

The v5 preloader counted 00 to 100 on a fixed one second cubic ease and lifted its curtain regardless of what had actually loaded. In v6 the count is a real load gate: it holds at 99 until `document.fonts.ready` and the window `load` event have both resolved, so the curtain never lifts onto a page that still shifts. A 3.5s `MAX_WAIT` cap races the gate so slow networks cannot trap anyone.

```
Promise.race([
  Promise.all([fontsReady, pageLoaded]),
  cap // setTimeout(resolve, 3500)
]).then(() => { assetsReady = true; });

if (!assetsReady) count = Math.min(count, 99);
```

When the curtain lifts (`translateY(-100%)`), `body.is-loaded` triggers the hero letter cascade, and the shader's ignition burst fires at the same moment. **Load, ignite, cascade: one choreographed instant.**

KEPT FROM V5

- **Film grain**, a fixed SVG `feTurbulence` data URI at 0.045 opacity over everything.
- **Scroll parallax**: `data-parallax` images translate and scale (1.12) against scroll.
- **Work list hover**: rows invert via a `scaleY::before` wipe, with a cursor following preview card at lerp 0.12.
- **Live IST clock**, `Intl.DateTimeFormat` in Asia/Kolkata, ticking every second.
- **Fullscreen mobile menu** with staggered links, right click guard, console signature, ASCII art in the source.

A NOTE ON THE COUNT

The gate is honest in both directions. If everything is already cached, which is the normal case once the app is installed, `assetsReady` resolves immediately and the count runs its full one second ease anyway. The curtain is a *promise about layout*, not a progress bar.

THE ORDER THINGS HAPPEN

Four scripts, no module system, no bundler, and therefore no dependency graph to resolve. Order is expressed the old way: by where the `<script>` tags sit, all of them at the end of `<body>`. The one flag that has to run early, `html.js`, is set inline in the `<head>` so the no JavaScript stylesheet never flashes.

| #  | MOMENT                 | WHAT HAPPENS                                                                                        |
|----|------------------------|-----------------------------------------------------------------------------------------------------|
| 01 | <head>                 | <code>html.js</code> is set inline; <code>fonts.css</code> and <code>style.css</code> are requested |
| 02 | <code>main.js</code>   | Preloader count starts; <code>fitLines()</code> measures the hero                                   |
| 03 | <code>motion.js</code> | Adds <code>html.has-motion</code> , splits the hero into letters, calls <code>__refitHero</code>    |
| 04 | <code>shader.js</code> | Compiles the program, mounts the canvas, or removes it and gives up                                 |

| #  | MOMENT                      | WHAT HAPPENS                                                              |
|----|-----------------------------|---------------------------------------------------------------------------|
| 05 | <code>fonts + load</code>   | Both resolve, or 3.5s passes. <code>assetsReady</code> flips              |
| 06 | <code>count hits 100</code> | 150ms pause, then <code>body.is-loaded</code>                             |
| 07 | <code>curtain lifts</code>  | Letter cascade runs and <code>uBurst</code> fires at 1.4, together        |
| 08 | <code>sw-register.js</code> | On <code>load</code> : registers, or tears down a worker on a dead origin |

## (08) · THE OFFLINE APPLICATION

# A ROOM WITH NO *internet*

The brief for this one was literal: in a room with no internet, the whole portfolio, the résumé, the certificates and **the type it is set in** have to already be on the phone. That is a stricter requirement than "has a service worker", and it changed four things.

## PRECACHED IN FULL, ATOMICALLY

The earlier worker filled its cache in as pages happened to be visited. This one declares a **SHELL** of everything needed to render all four pages with the network switched off, and installs it with a single `cache.addAll()`. That call is all or nothing on purpose: **if one entry is wrong the install fails loudly**, rather than leaving a half usable app that only breaks once you are already offline.

| TIER   | CONTENTS                                                                                              |
|--------|-------------------------------------------------------------------------------------------------------|
| SHELL  | 4 pages, 2 stylesheets, 7 scripts, manifest, icons, 3 PDFs, 16 font files. Atomic.                    |
| MEDIA  | Certificates and feature photographs. Added one at a time, allowed to fail.                           |
| Thumbs | Read from <code>photos.json</code> at install, so adding a photograph never means editing the worker. |

Full size photographs are deliberately left out. They are roughly 40 MB, and they are cached individually as they are opened instead.

## TYPE IS PART OF THE SHELL

All sixteen woff2 subsets ship in **SHELL**. This is why the fonts were brought in house in the first place: an offline site that silently falls back to Arial is not *this* site.

## TWO STRATEGIES, CHOSEN PER DESTINATION

**Network first** for navigations and for scripts and styles, so the newest deploy wins whenever there is a network and the cache is only what is left when there is not. **Cache first** for fonts, images and PDFs, which cannot change without a redeploy, and a redeploy changes the cache name anyway.

```
if (request.method !== 'GET') return; // the form's POST
if (new URL(request.url).origin !== self.location.origin) return; // third party

const isNavigation = request.mode === 'navigate'
|| request.destination === 'document';
const isCode = request.destination === 'script'
|| request.destination === 'style';

if (isNavigation || isCode)
  return event.respondWith(networkFirst(request, isNavigation));
event.respondWith(cacheFirst(request));
```

An offline deep link to a page that was never cached falls back to `/index.html`, so a visitor gets the portfolio rather than the browser's dinosaur. The contact form's POST and every cross origin request are left entirely alone.

## DEAD ORIGIN EVICTION

`sw-register.js` registers only on `mosambiswas.com` and on `localhost`, which is treated as production so the offline behaviour can be tested before it ships. **Anywhere else**, any previously registered worker is unregistered and every cache deleted. That eviction is not theoretical: it is what emptied the old `mosambiswas.me` app when the domain lapsed.

## THE ONE DEPLOY RULE

**VERSION** at the top of the worker is bumped on **every** deploy. Currently **v18**. The name change is the entire eviction mechanism; forget it and visitors keep the old site.

(09) · RÉSUMÉ, CONTACT, PROOF

# BUILT FOR THE PERSON WHO HAS *four minutes*

A friend reviewed the site and the verdict was blunt: it was beautiful and it was slow to answer the only question a recruiter asks. This section is what came out of that review.

## RESUME.HTML

A web résumé in the site's own design system, reusing `style.css`, `shader.js` and `main.js` with page specific styles inline. Five sections: where I've *worked*, six of *twenty plus*, published and *presented*, the paper *trail*, and contact.

It carries a `@media print` stylesheet that strips to clean black on white, plus two downloads: the colour PDF and a black and white one. The nav gets an accent pill CTA on desktop and 06 · Résumé in the mobile menu.

## THE CONTACT FORM THAT WILL NOT LIE

LET'S TALK is a `role="button"` heading that morphs into a name, email and message form in place, closable with Escape. It posts by `fetch` to FormSubmit's AJAX endpoint with a honeypot field and no captcha.

### The bug worth recording.

FormSubmit answers 200 even when it refuses a message, and says so only in the body. Branching on `res.ok` alone showed visitors "Sent ✓" for mail that never left.

```
const body = await
res.json().catch(() => null);
if (!body || String(body.success)
!= 'true')
  throw new Error(body?.message
|| 'rejected');
```

Failure is not silent either: it points at the email address below, which is **click to copy** via the Clipboard API, falling back to `mailto:` when the API is missing or throws.

## WORK ROWS CARRY OUTCOMES

Hovering a project no longer shows a name tile. It shows an **outcome stat card**: the index and year, one metric taken from the résumé, and a single line of what it did. The numbers are held to the résumé, so the two can never drift apart.

Where a project is deployed, the row links to the *running app* rather than the repository, and the title carries a **Live** badge. The title is the only grid column that survives every breakpoint, which is why the badge lives there.

## TEN ROWS, ONE NEW

01 · AMINAL leads the list: a local first voice assistant for Windows, Python with Whisper and Ollama, documented in its own public repository. Nothing in it leaves the machine.

## THE TEN CARDS ONE NUMBER AND ONE LINE EACH, EVERY NUMBER TRACED TO THE RÉSUMÉ

| PROJECT    | STAT  | OUTCOME                                       |
|------------|-------|-----------------------------------------------|
| AMINAL     | Local | Voice, vision and memory on your own GPU      |
| Sim Sim    | 70%   | Faster task execution, voice plus a GUI       |
| Rail Mitra | Zero  | Conflict schedules, by constraint programming |
| MindFirst  | 95%   | Screening accuracy, 100+ students             |
| PixelForge | 0.90  | ROC-AUC, leukemia detection                   |

| PROJECT           | STAT | OUTCOME                                |
|-------------------|------|----------------------------------------|
| IntelliStatement  | LLM  | Extraction from PDFs and images        |
| Committee Connect | Live | Real time Firebase sync                |
| PixelShift        | Zero | Uploads. All conversion in the browser |
| Flex Card         | PNG  | Share ready hardware spec cards        |
| To-Do             | E2E  | Task boards with auth, the whole loop  |

(10) · CERTIFICATES: THE DOSSIER

# PROOF OF WORK, *declassified*

The certificates page gets its own motion language, paper, stamps and filing cabinets, gated behind `html.has-dossier`. Three featured credentials, eight in the archive.

## REDACTION BAR HERO

The title **PROOF OF WORK** arrives like a declassified document: solid bars, ink for the first line, vermilion for the outlined second, sweep across each line (`scaleX` 0 to 1 from the left, then 1 to 0 from the right), and the text snaps visible mid sweep, exactly when the bar covers it.

```
@keyframes dossier-sweep {
  0%   { transform: scaleX(0); origin: 0 50%; }
  46%  { transform: scaleX(1); origin: 0 50%; }
  54%  { transform: scaleX(1); origin: 100% 50%; }
  100% { transform: scaleX(0); origin: 100% 50%; }
}
```

## STAMPED FEATURES

Featured certificates land with an overshoot bezier (0.2, 0.9, 0.3, 1.15) from `scale(1.05)` `rotate(0.6deg)`, stamped onto the page, followed by a decaying vermilion stamp flash `box-shadow`. The modal image reuses the same stamp in.

## FILED TILES AND TYPED LABELS

Archive tiles file in one by one, 55ms stagger, each with a slight index derived rotation (`--i - 3`)  $\times 0.35\text{deg}$ , like papers squared off by hand. Section labels type themselves character by character with a `█` caret at 26ms per glyph; the "08 certificates" aside counts itself up. All triggered by `IntersectionObservers`.

## PAPER TILT AND SHEEN

On desktop, every certificate tilts in 3D under the cursor, `perspective(720px)` with up to 9°/7° of rotation, while a radial sheen follows the pointer via `--gx` and `--gy` custom properties, like *light raking across a glossy print*.

```
e1.style.setProperty('--ry',
  `${((px - 0.5) * 9).toFixed(2)}deg`);
e1.style.setProperty('--rx',
  `${((0.5 - py) * 7).toFixed(2)}deg`);
```

## WHAT IS IN THE CABINET

Three credentials are featured with their own panel and a stamp in animation. The remaining eight are archive tiles, filed in sequence.

### THE BIG THREE

- 01 Research presentation, ICAIHI 2025
- 02 Software development internship, Sedna Technologies
- 03 Samsung Innovation Campus, AI

### EVERYTHING ELSE

- 01 Android Compose Camp05 Industry focused Java workshop
- 02 AWS Machine Learning06 Programmable circuit building
- 03 Generative AI07 Hacktoberfest
- 04 Agile and Scrum fundamentals08 Smart India Hackathon 2025

(11) · SHEICHOBI: THE DARKROOM

# PRINTS THAT DEVELOP AS YOU *watch*

**Shei Chobi**, Bengali for "that picture", is the photography wing: 84 photographs in six collections, and in v6 it behaves like a darkroom, light, chemistry and optics, gated behind `html.has-darkroom`.

## THE DEVELOPING BATH

The hero title **SHEI CHOBI** arrives overexposed, `blur(18px)` `brightness(2.3)`, and settles into focus over 1.7s, like a print in developer fluid. Every archive photo goes through the same bath: from `blur(14px)` `brightness(1.9)` `grayscale(1)` to full colour, batch staggered 90ms. Because the grid is rebuilt on every filter and load more, a `MutationObserver` re-watches new cards automatically.

## APERTURE LIGHTBOX

Clicking a photo opens the lightbox as an aperture: a `clip-path: circle()` expands from 0% to 141% centred on the exact click point, captured into `--cx` and `--cy`. Between photos, a shutter blink, brightness flash and blur, restarted by forcing reflow when the image `src` mutates.

## FOCUS PULL

Hovering the contact sheet racks focus like a lens: the hovered frame stays sharp while every sibling falls to `blur(3px)` `brightness(0.75)`. A pure CSS `:hover` chain, desktop only.

### REVISED · WHAT THE LAG TAUGHT

The grain originally breathed by applying a **filter to the sheet itself**, up to 2.6px of motion blur. That forces the browser to re-rasterize the entire grid every frame, and scrolling janked on exactly the devices that matter. The loop now animates **opacity only**, and it parks itself the moment velocity settles instead of running forever.

## THUMBNAILS, NOT ORIGINALS

The second half of the fix was weight. The archive grid and the collection previews load 900px wide webp thumbnails from

`Photographs/thumbs/`; only the lightbox loads the original. A small Node script regenerates them, and a photograph without a thumbnail still works, it just loads slower until the script is run.

```
// generate-thumbs.js · run after
adding photos
await sharp(src)
  .resize({ width: 900,
    withoutEnlargement: true })
  .webp({ quality: 82 })
  .toFile(dest);
```

The same `photos.json` that drives the grid is what the service worker reads at install time to decide which thumbnails to precache.

*"Every page speaks the same design language, but each one moves in **its own dialect**: embers, dossiers, darkrooms."*

(12) · ENGINEERING LEDGER

# FAST, ACCESSIBLE, *honest*

## PERFORMANCE DECISIONS

- **Zero dependencies.** No framework tax; the whole site is 2,192 lines of vanilla JS.
- **Transform and opacity only.** Every animation runs on compositor friendly properties; staggers via `--i` custom properties, not per element JS timers.
- **One rAF loop per system.** Motion, shader and darkroom each run a single frame loop; observers do the triggering, and the darkroom loop parks itself when idle.
- **Adaptive shader quality,** 0.5× render scale, low power context, self degrading on slow frames.
- **Thumbnails in the grid,** originals only in the lightbox.
- **Preloader as load gate,** no layout shift after the curtain.

## ACCESSIBILITY & RESILIENCE

- **Reduced motion.** All four v6 engines bail on `prefers-reduced-motion`; the shader canvas removes itself; CSS gates make the v6 styles inert without JS.
- **No JS fallback.** Reveals are gated behind `html.js`; content is fully readable without a single script.
- **Skip link, sr-only h1, semantic landmarks, keyboard reachable modals and form.**
- **Honest status.** The contact form reports what actually happened, and points at a copyable address when it fails.

## SEO & DISTRIBUTION

- **JSON-LD,** WebSite and Person structured data, OG and Twitter cards, sitemap and robots.txt.
- **Installable,** manifest with maskable icon, screenshots and three shortcuts: Résumé, Certificates, Photography.
- **CNAME** to `www.mosambiswas.com`, canonical redirects for the gallery.

## CONTENTS OF RECORD

- 10 Projects: AMINAL, Sim Sim, Rail Mitra, MindFirst, PixelForge, IntelliStatement, Committee Connect, PixelShift, Flex Card, To-Do. Two are live apps.
- 11 Certificates, three featured and eight archived.
- 84 Photographs in six collections.
- 01 IEEE publication: psychometric ML screening, ICAIHI 2025.  
DOI  
10.1109/ICAIIHI67124.2025.11403838
- 04 Pages, all of them available offline.

(•) · CONTACT

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(13) · RECOGNITION

# OUT IN THE *wild*

On 10 July 2026 the site was submitted to **Dark Mode Design** (darkmodedesign.com), a curated directory of sites built dark by default, not dark mode as a toggle. The submission note doubled as a one paragraph brief of the entire project, reproduced here exactly as sent, em dashes and all:

*"Warm-black editorial design — #0b0b0a with a single vermilion accent — hand-built with zero frameworks. Behind every page runs a hand-written GLSL ember field; the custom cursor whispers captions on individual words and turns into a lens that reveals the page's hidden subtext. Full dark from the first pixel — **dark mode isn't a toggle here, it's the material.**"*

SUBMISSION

|              |                                                                              |
|--------------|------------------------------------------------------------------------------|
| Directory    | Dark Mode Design, darkmodedesign.com                                         |
| Submitted    | 10 Jul 2026                                                                  |
| Pitch covers | Zero framework build, hand written GLSL ember field, whisper and lens cursor |

OUTCOME

|             |                                                                  |
|-------------|------------------------------------------------------------------|
| Status      | Accepted                                                         |
| Reviewer    | Cai Cardenas                                                     |
| Queued live | Within about a week of acceptance, response received 20 Jul 2026 |

The pitch and the product turned out to be the same file: every claim in that one paragraph is documented, page by page, in the sections before this one. The only thing the paragraph did not promise is the one this revision adds, which is that all of it *still works with the network switched off*.

DOCUMENT HISTORY

| EDITION            | COVERS                                                                                                   |
|--------------------|----------------------------------------------------------------------------------------------------------|
| Vol. 06 · Jul 2026 | Ember field, motion system, dossier and darkroom. Eleven sections.                                       |
| Rev. · Sep 2026    | Offline application, résumé and contact surface, thumbnail pipeline, whisper scoping. Thirteen sections. |

HOW TO VERIFY ANY CLAIM HERE

Every number in this document was read out of the repository rather than remembered. The shader constants are in `shader.js`, the cache tiers in `service-worker.js`, the stat cards in `index.html`. Nothing is minified, so **view source is the primary source**.